Fundamentals of Linear Control:
A Concise Approach

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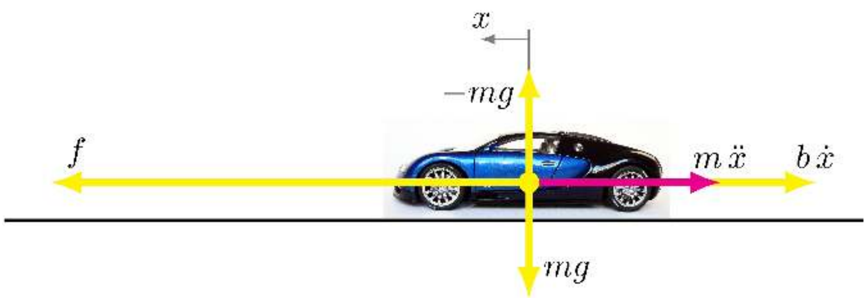
Chapter 2: Dynamic Systems

linearcontrol.info/fundamentals

2.1 Dynamic Models

Longitudinal car model



Newton's law

$$m\,\dot{v}(t) + b\,v(t) = f(t)$$

Force model

$$f(t) = p\,u(t)$$

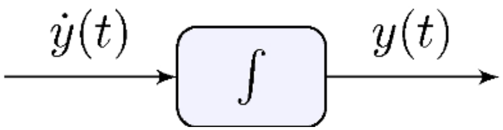
Ordinary differential equation

$$\dot{y}(t) + \frac{b}{m}\,y(t) = \frac{p}{m}u(t)$$

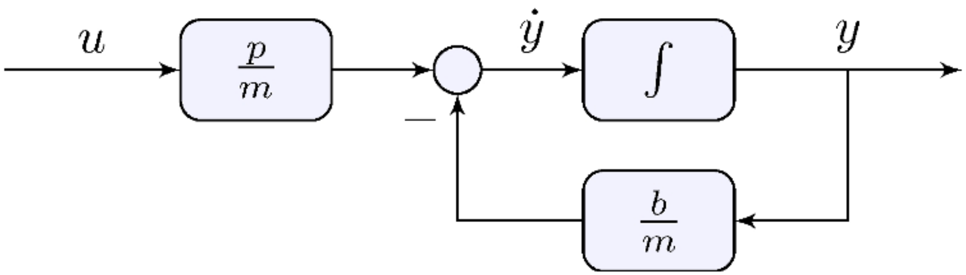
- input: u is pedal excursion in inches
- output: y is car velocity in mph

2.2 Block-Diagrams for Differential Equations

Basic element: integrator



Longitudinal car model



$$\dot{y}(t) = \frac{p}{m}u(t) - \frac{b}{m}y(t)$$

Note the presence of feedback!

2.3 Dynamic Response

Longitudinal car model

$$\dot{y}(t) + \frac{b}{m}\, y(t) = \frac{p}{m} u(t)$$

Constant pedal intput

$$u(t) = \tilde{u}, \quad t \geq 0$$

Particular solution

$$y_P(t) = \tilde{y} = \left(\frac{p}{m} \middle/ \frac{b}{m} \right) \tilde{u} = \frac{p}{b} \tilde{u}$$

Longitudinal car model: homogenous equation

$$\dot{y}(t) + \frac{b}{m}\, y(t) = 0$$

Homogenous solution

$$y_H(t) = e^{\lambda t}$$

Characteristic equation

$$\dot{y}_H(t) + \frac{b}{m} y_H(t) = \left(\lambda + \frac{b}{m}\right) e^{\lambda t} = 0 \qquad \Longrightarrow \qquad \lambda + \frac{b}{m} = 0$$

Longitudinal car model with constant pedal input

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} \tilde{u}$$

Particular + homogenous solution

$$y(t) = y_P(t) + \beta\, y_H(t) = \tilde{y} + \beta\, e^{\lambda t}$$

Initial conditions

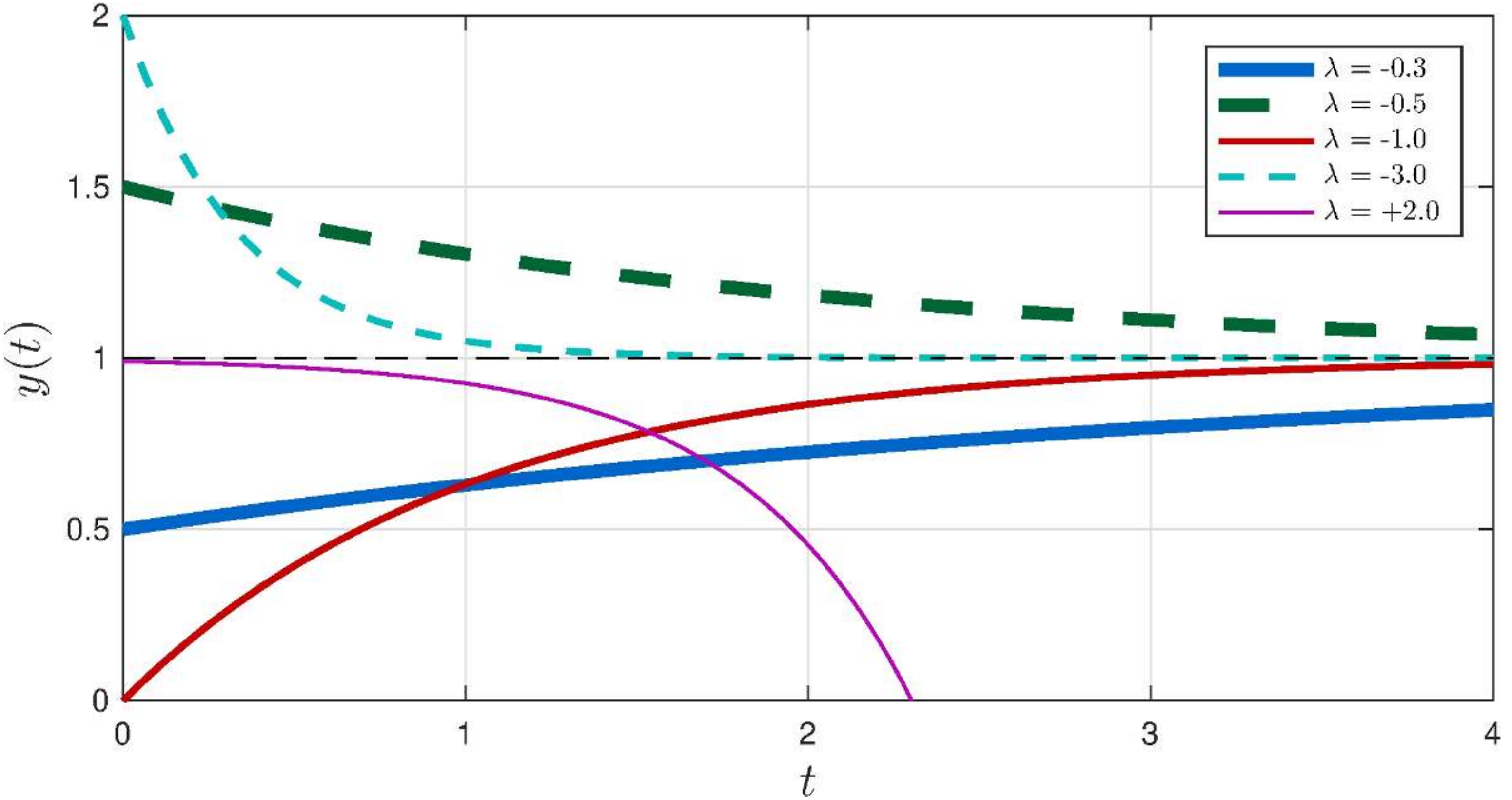
$$y(0) = \tilde{y} + \beta = y_0 \qquad \Longrightarrow \qquad \beta = y_0 - \tilde{y}$$

Complete solution

$$y(t) = \tilde{y}\left(1 - e^{\lambda t}\right) + y_0\, e^{\lambda t}, \qquad \lambda = -\frac{b}{m}, \qquad \tilde{y} = \frac{p}{b}\tilde{u}$$

Examples

$$y(t) = \tilde{y}\left(1 - e^{\lambda t}\right) + y_0\, e^{\lambda t}, \qquad \tilde{y} = 1$$



Time-constant

Definition

$$\tau = \{t : y(t) = (1 - e^{-1})\tilde{y} \approx 0.63\,\tilde{y}\}$$

First-order systems

$$\tau = -\frac{1}{\lambda} \qquad y(\tau) \approx 0.63\,\tilde{y}, \quad y(3\tau) \approx 0.95\,\tilde{y}$$

Rise-time

Definition

$$t_r = t_2 - t_1, \qquad t_1 = \{t : y(t) = 0.1\,\tilde{y}\}, \qquad t_2 = \{t : y(t) = 0.9\,\tilde{y}\}$$

First-order systems

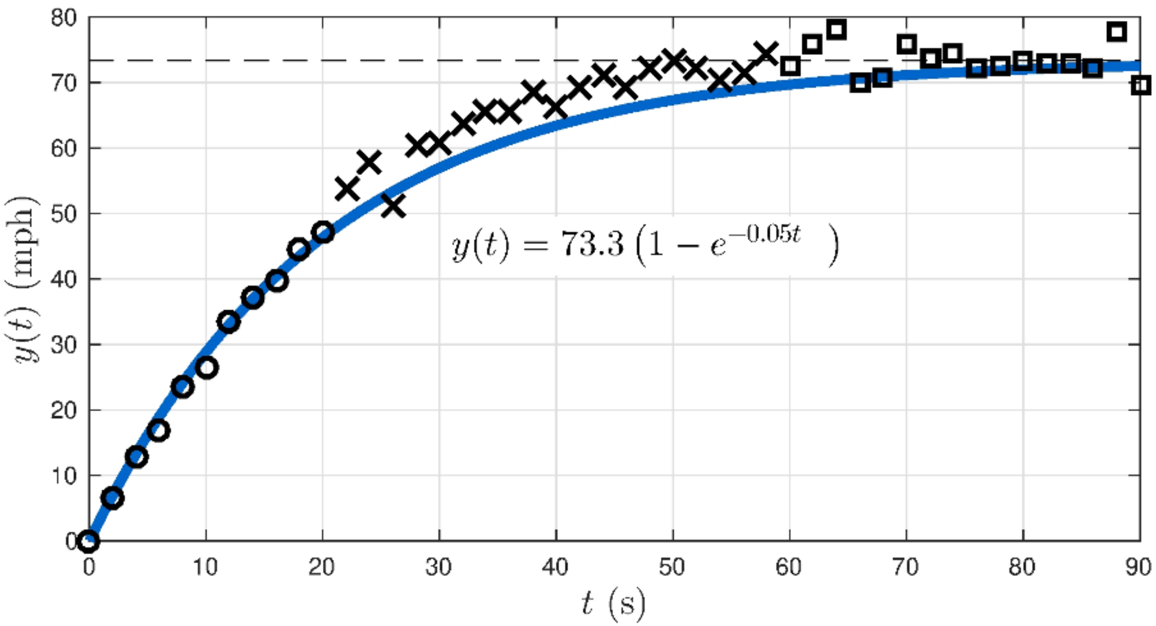
$$t_r = \ln(1/9)\,\lambda^{-1} = \ln(9)\,\tau \approx 2.2\,\tau$$

2.4 Experimental Dynamic Response

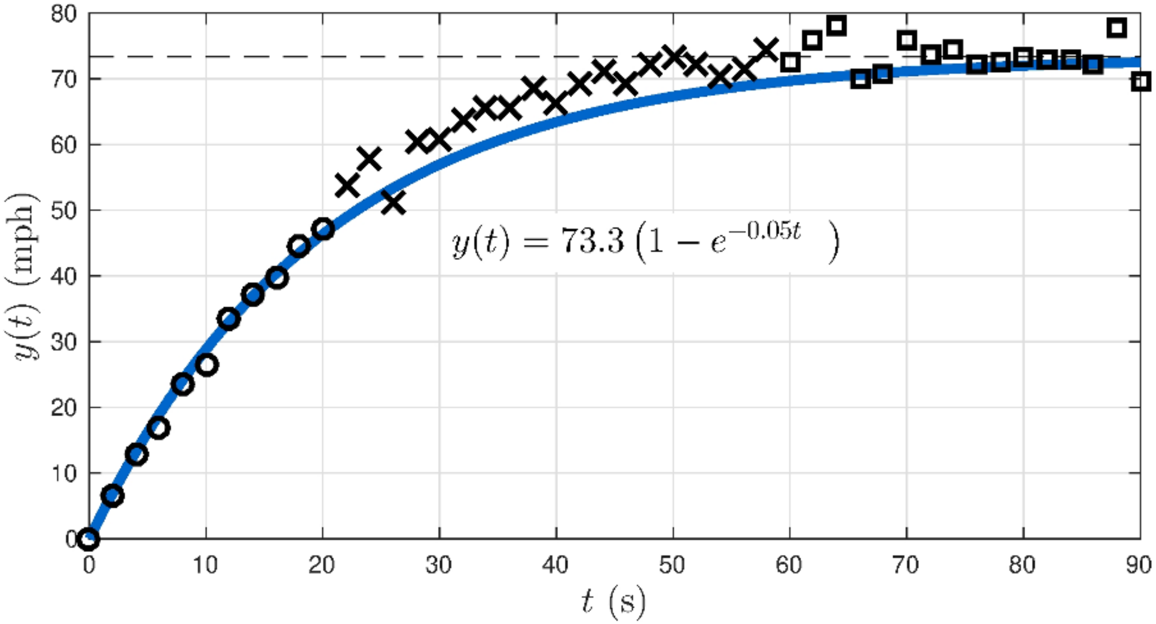
Experiment

Zero initial velocity

Constant pedal excursion: $\tilde{u} = 1$ in



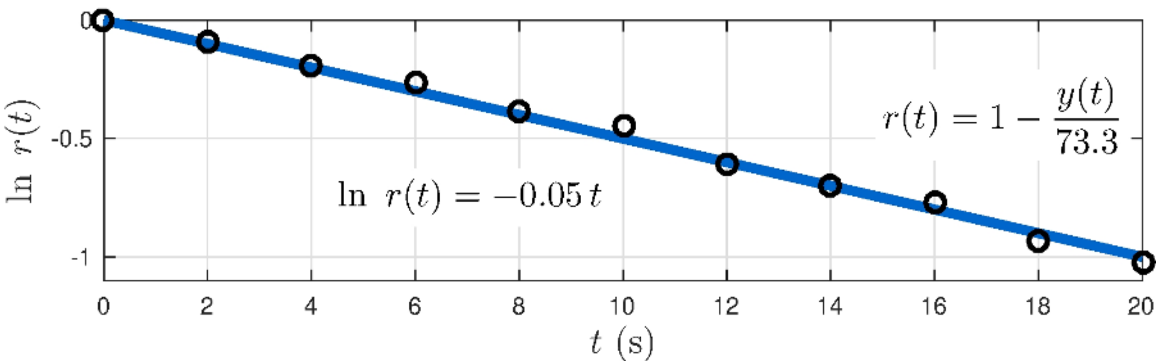
Model fitting



Squares: steady-state solution $\implies$ $\tilde{y} \approx 73.3$ mph

Model fitting

$$r(t) = 1 - y(t)/\tilde{y} = e^{\lambda t} \qquad \Longrightarrow \qquad \ln r(t) = \lambda t$$



Circles: $\lambda \approx -0.05\ \mathrm{s}^{-1}$

Model fitting

$$
\begin{aligned}
\frac{b}{m} &= -\lambda \approx 0.05\,\mathrm{s}^{-1}, \\
\frac{p}{b} &= \frac{\tilde{y}}{\tilde{u}} \approx 73.3\,\mathrm{mph/in}, \\
\frac{p}{m} &= \frac{b}{m} \times \frac{p}{b} = -\lambda \times \frac{\tilde{y}}{\tilde{u}} = 3.7\,\mathrm{mph/(in\ s)}
\end{aligned}
$$

Experimental model

$$
\dot{y}(t) + 0.05\,y(t) = 3.7u(t)
$$

Be careful

Experiment was conducted for $u(t) = \tilde{u}$ constant but we *wish* model to hold for general $u(t)$!

2.5 Dynamic Feedback Control

Car model

Dynamic model

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t)$$

Feedback controller

$$u(t) = K\, e(t), \qquad e(t) = \bar{y} - y(t)$$

Closed-loop system

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) = \frac{p}{m} K \bar{y}$$

Closed-loop response

Dynamic model

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m}K\right)y(t) = \frac{p}{m}K\bar{y}, \qquad y(0) = y_0, \qquad \lambda = -\frac{b}{m} - \frac{p}{m}K$$

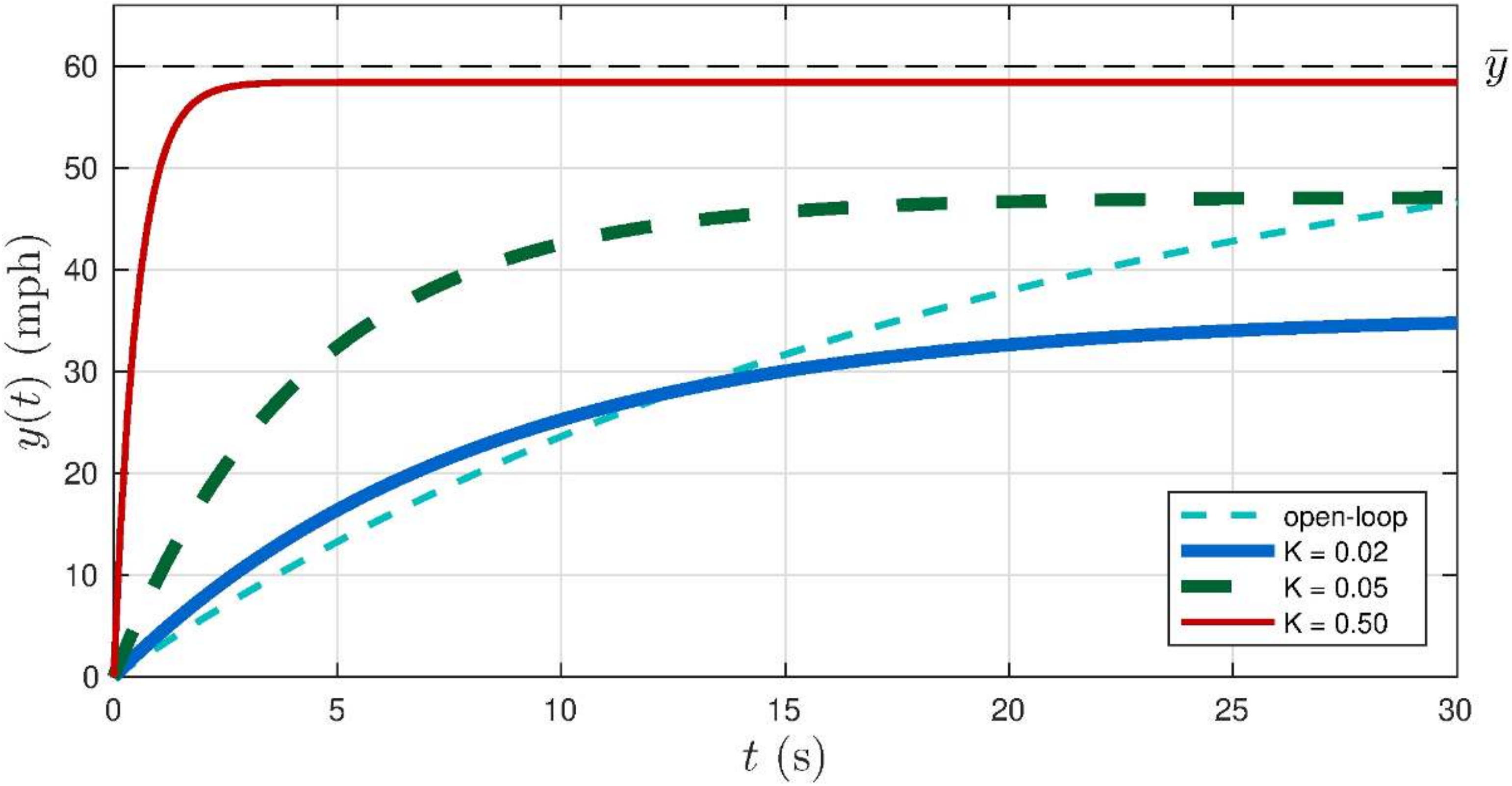
Output

$$y(t) = \tilde{y}\left(1 - e^{\lambda t}\right) + y_0\, e^{\lambda t}, \qquad \tilde{y} = \frac{\frac{p}{m}K}{\frac{b}{m} + \frac{p}{m}K}\bar{y} = \frac{\frac{p}{b}K}{1 + \frac{p}{b}K}\bar{y}$$

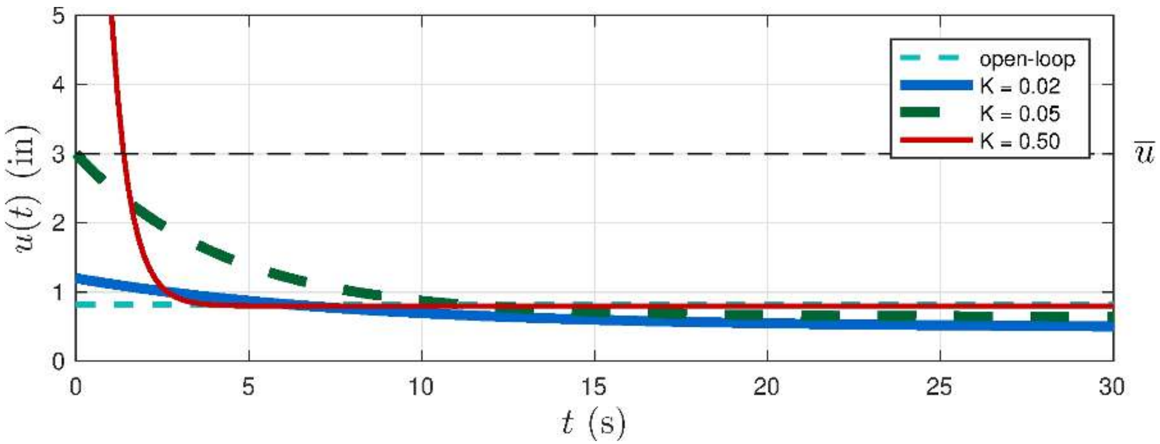
Time-constant

$$\lambda = -\frac{b}{m} - \frac{p}{m}K, \qquad \tau = -\lambda^{-1} = \frac{m}{b + pK}$$

Output


60
50
40
30
20
10
0
0
5
10
15
20
25
30
$\bar{y}$
$y(t)$ (mph)
t (s)
open-loop
K = 0.02
K = 0.05
K = 0.50

Control



Largest control at $t = 0$

$$u(0) = Ke(0) = K(\bar{y} - y(0)) = K\,\bar{y}$$

Saturation

$$\bar{y} = 60 \text{ mph} \implies u(t) > \bar{u} = 3 \text{ in for } K > 0.05!$$

Steady-state

$$\lim_{t\to\infty} y(t) = \tilde{y} = H(0)\,\bar{y}, \qquad H(0) = \frac{\frac{p}{b}K}{1+\frac{p}{b}K}$$

$$\lim_{t\to\infty} e(t) = \bar{y} - \tilde{y} = S(0)\,\bar{y}, \qquad S(0) = \frac{1}{1+\frac{p}{b}K}$$

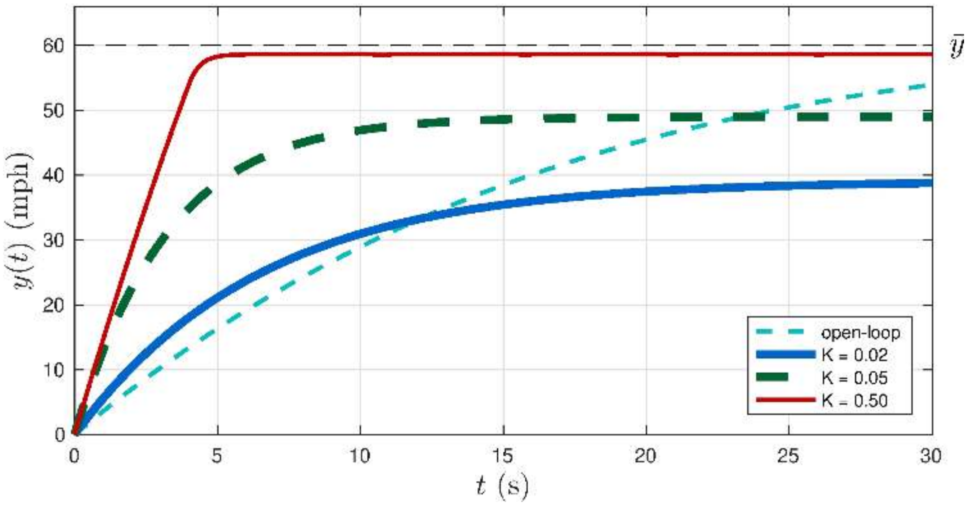
Effect of the gain

K	$H(0)$	$S(0)$	$\tilde{y}$ (mph)	τ (s)	t_r (s)
open-loop	1.00	0.00	60	20.0	43.9
0.02	0.60	0.40	36	8.1	17.8
0.05	0.79	0.21	47	4.3	9.4
0.50	0.97	0.03	58	0.5	1.2

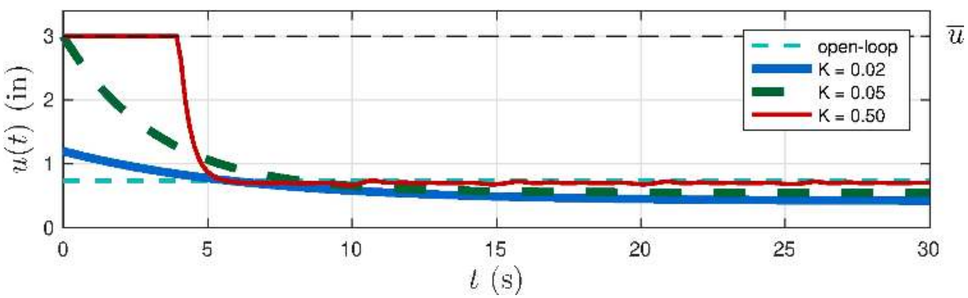
$\bar{y} = 60$ mph

2.6 Nonlinear Models

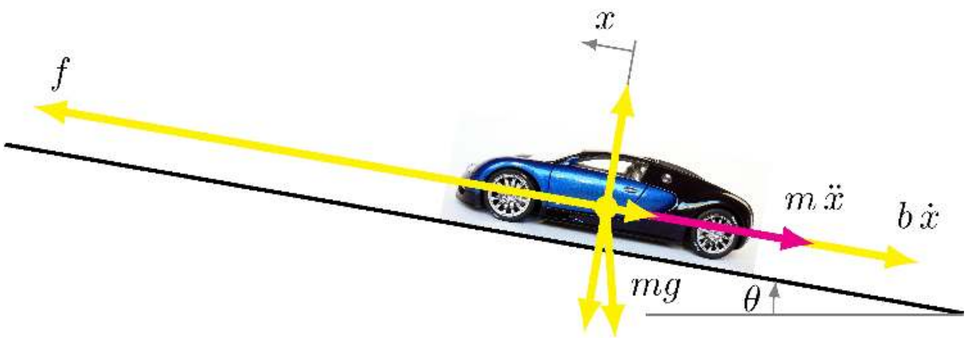
Output is rate-limited



Control saturates



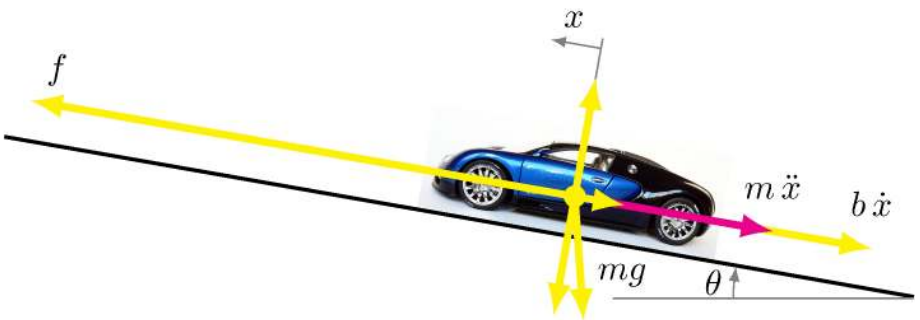
2.7 Disturbance Rejection

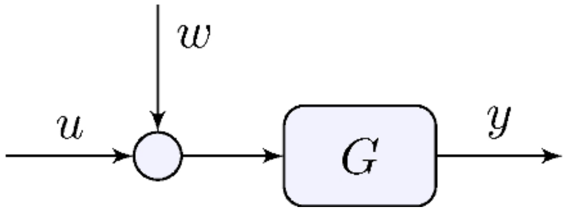


Car on inclined plane model

$$\dot{y}(t) + \frac{b}{m}\,y(t) = \frac{p}{m}u(t) - g\sin(\theta(t))$$

Disturbance Model





Car on inclined plane with disturbance

$$\dot{y}(t) + \frac{b}{m}\,y(t) = \frac{p}{m}(u(t) + w(t)), \qquad w(t) = -\frac{mg}{p}\sin(\theta(t))$$

Car with disturbance

$$\dot{y}(t) + \frac{b}{m}\, y(t) = \frac{p}{m}(u(t) + w(t))$$

Constant disturbance

$$w(t) = \bar{w}, \quad t \geq 0$$

Feedback controller

$$u(t) = K\, e(t), \qquad e(t) = \bar{y} - y(t)$$

Impact of disturbance on steady-state

Model

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m}K\right)y(t) = \frac{p}{m}(K\bar{y} + \bar{w}), \quad y_0 = H(0)\bar{y}$$

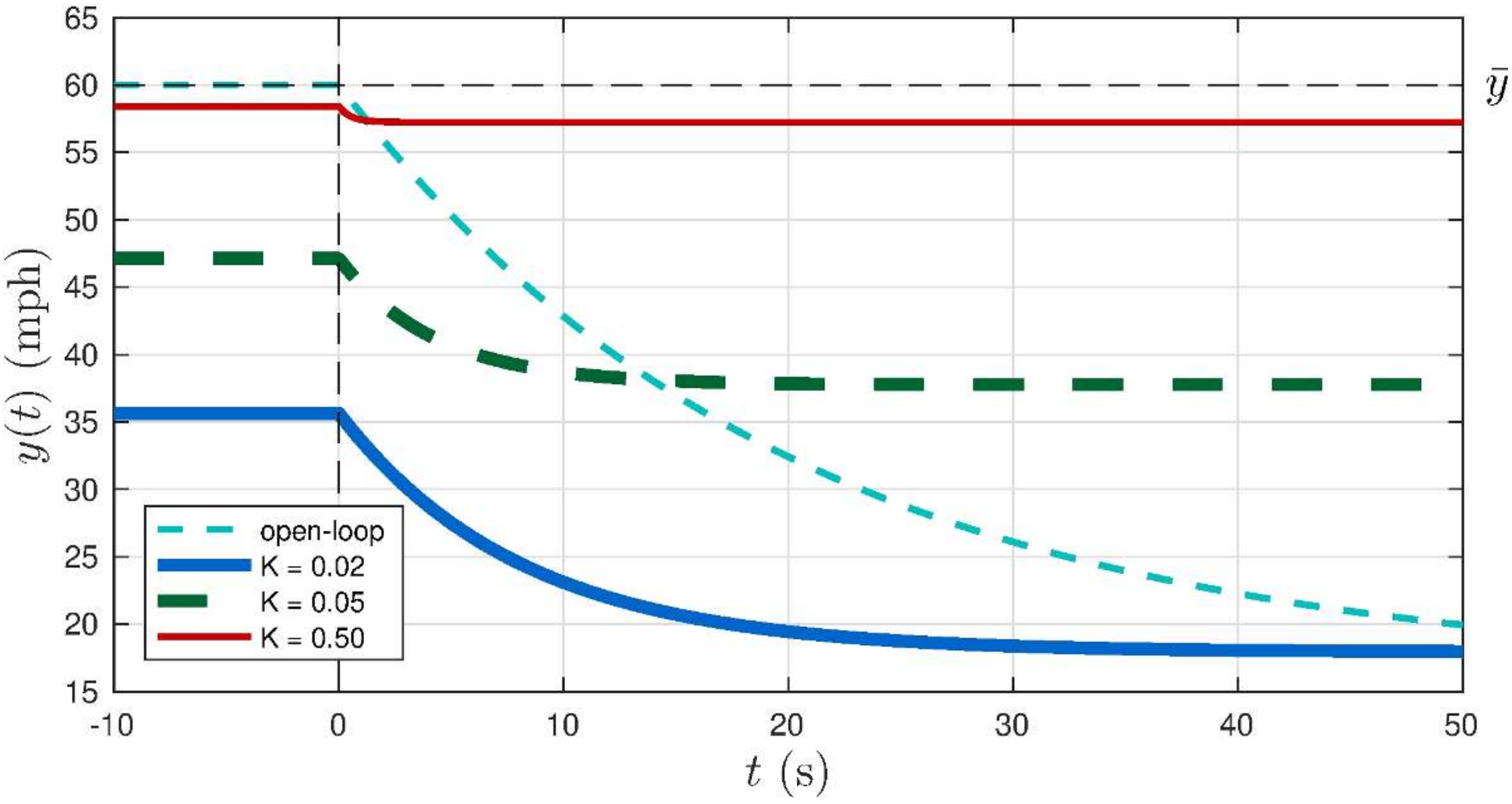
Output

$$y(t) = H(0)\,\bar{y} + \left(1 - e^{\lambda t}\right)D(0)\,\bar{w}, \qquad H(0) = \frac{\frac{p}{b}K}{1 + \frac{p}{b}K}, \qquad D(0) = \frac{p}{b}\,\frac{1}{1 + \frac{p}{b}K}$$

Deviation from steady-state

$$\Delta y(t) = y(t) - H(0)\bar{y} = \left(1 - e^{\lambda t}\right)D(0)\,\bar{w}$$

Output



65
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$\bar{y}$
$y(t)$ (mph)
t (s)
open-loop
K = 0.02
K = 0.05
K = 0.50

2.8 Integral Action

Toilet tank model

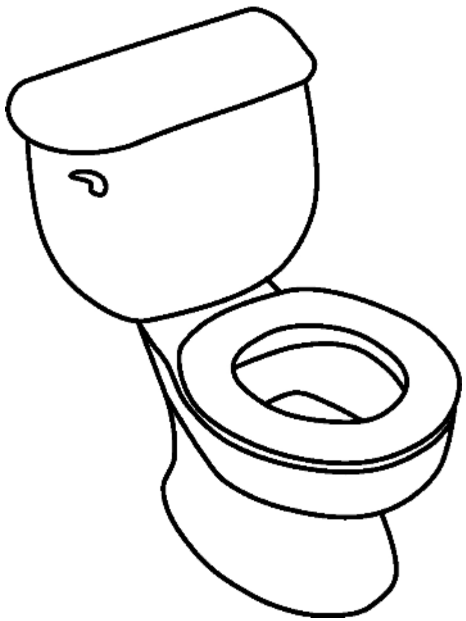
$$\dot{y}(t) = \frac{1}{A} u(t)$$

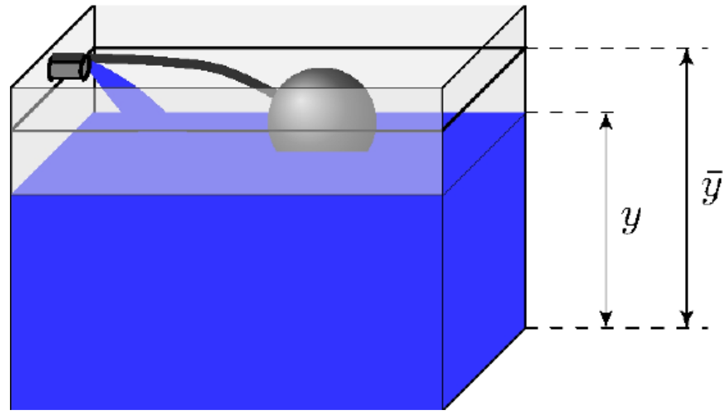
Ballcock valve

$$u(t) = K(\bar{y} - y)$$

Closed-loop model

$$\dot{y}(t) + \frac{K}{A} y(t) = \frac{K}{A} \bar{y}(t)$$





Closed-loop response

Model

$$\dot{y}(t) + \frac{K}{A} y(t) = \frac{K}{A} \bar{y}(t), \qquad y(0) = y_0, \qquad \lambda = -\frac{K}{A}$$

Output

$$y(t) = \tilde{y}\left(1 - e^{\lambda t}\right) + y_0 \, e^{\lambda t}, \qquad \tilde{y} = \bar{y}$$

Steady-state

$$\lim_{t \to \infty} y(t) = \tilde{y} = H(0)\,\bar{y}, \qquad H(0) = 1$$
$$\lim_{t \to \infty} e(t) = \bar{y} - \tilde{y} = S(0)\,\bar{y}, \qquad S(0) = 0$$

Integral action $\implies$ zero tracking error!

Can we replicate that result in a more general setting? Answer is in Chapter 4

Learn more

- Book website